



INTEGRATED ELECTRONICS # 04

Design and analyze the behavior of N-MOS switch, & N-MOS as Inverter.

Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	DESIGN AND CODING	OBSERVATION/ RESULTS	SCORE

Performance:

Objective: To be familiar with the NMOS operation as switch, and inverter.

Apparatus:

- N-MOS transistors
- Resistors
- Multimeter
- Connecting wire
- LED

Theoretical Background:

N-MOSFET Operation:

In order for drain current flow from drain to source in an N-channel MOSFET, the gate to source voltage must be greater than the specific device voltage(V_T), and the drain to source voltage must be greater than zero. The threshold voltage is dependent upon the physical dimensions and parameters for the MOSFET device. For enhancement only N-MOSFET transistor the threshold voltage is positive, and the enhancement depletion, the threshold voltage is negative.

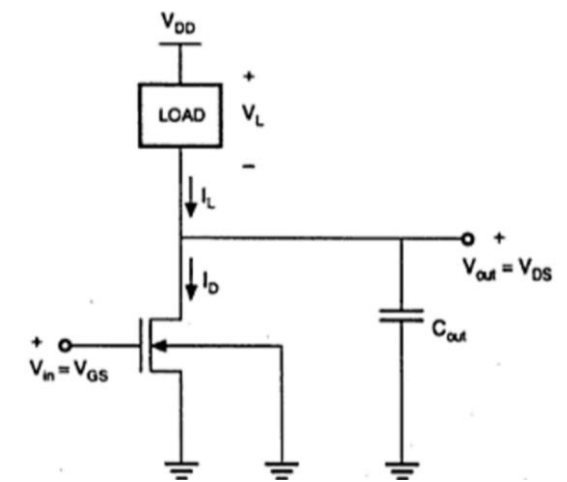
DISCUSSION:

The MOSFET was invented by Egyptian engineer Mohamed M. Atalla and Korean engineer Dawon Kahng at Bell Labs in 1959, and demonstrated in 1960.[1] They fabricated both PMOS and NMOS devices with a 20 μm process. However, the NMOS devices were impractical, and only the PMOS type were practical devices.[2]

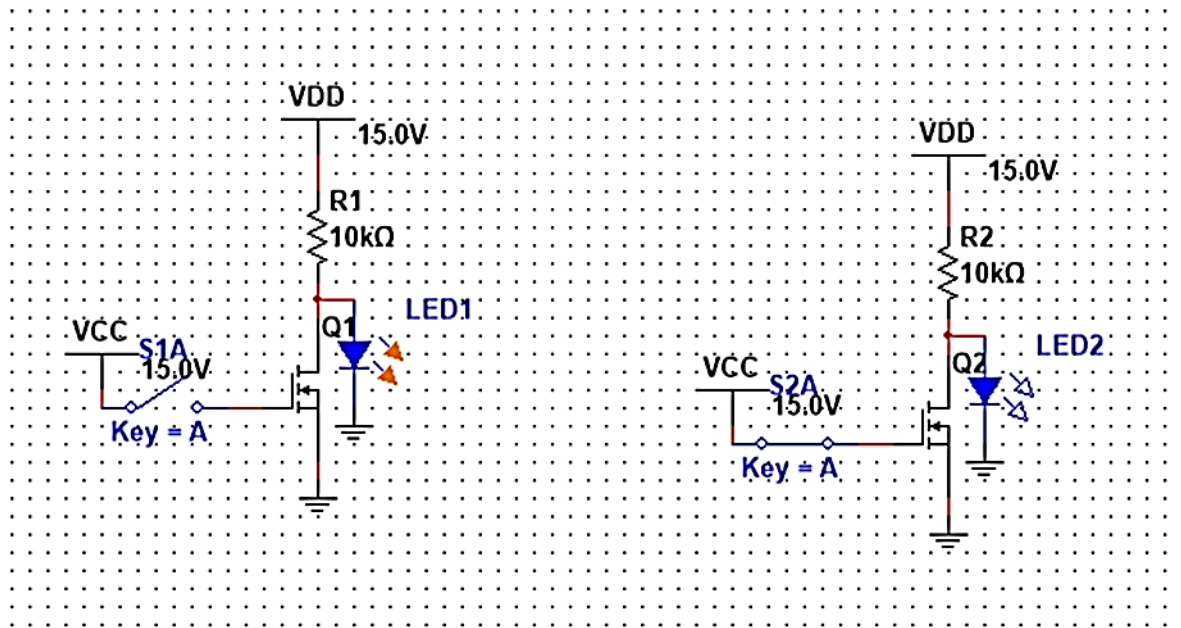
Fairchild Semiconductor fabricated several NMOS devices with channel lengths between 8 μm and 65 μm . [3]

N-type metal-oxide-semiconductor logic uses n-type MOSFETs to implement logic gates and other digital circuits. These NMOS transistors operate by creating an inversion layer in a p-type transistor body. This inversion layer, called the n-channel, can conduct electrons between n-type "source" and "drain" terminals. The n-channel is created by applying voltage to the third terminal, called the gate. Like other MOSFETs, NMOS transistors have four modes of operation: cut-off (or subthreshold), triode, saturation (sometimes called active), and velocity saturation.

The generalized circuit structure of an NMOS inverter is shown in the figure below. From the given figure, we can see that the input voltage of inverter is equal to the gate to source voltage of NMOS transistor and output voltage of inverter is equal to drain to source voltage of NMOS



SIMULATION:



Review Questions

1. Why mos inverter is distinguish inverter?

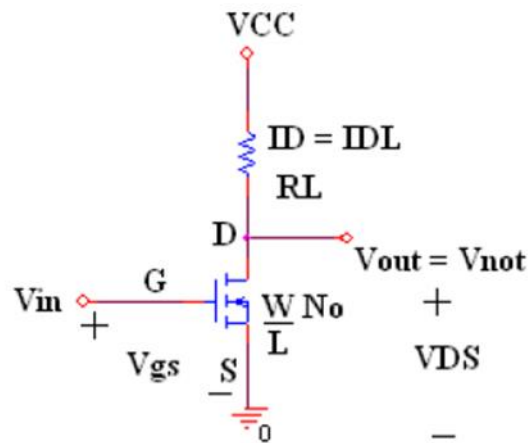
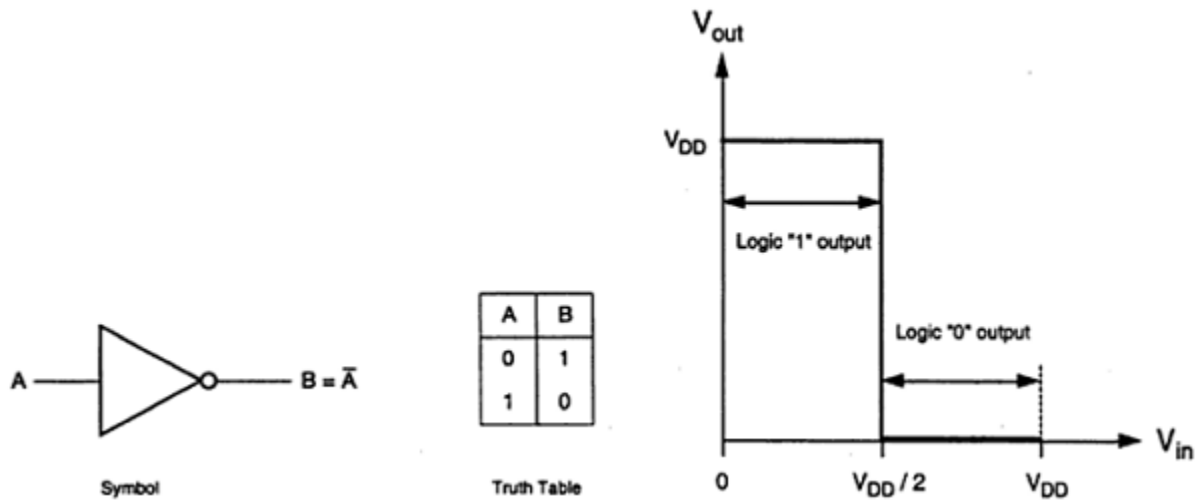
The inverter is truly the nucleus of all digital designs. Once its operation and properties are clearly understood, designing more intricate structures such as NAND gates, adders, multipliers, and microprocessors is greatly simplified. The electrical behavior of these complex circuits can be almost completely derived by extrapolating the results obtained for inverters.

The analysis of inverters can be extended to explain the behavior of more complex gates such as NAND, NOR, or XOR, which in turn form the building blocks for modules such as multipliers and processors. In this chapter, we focus on one single incarnation of the inverter gate, being the static CMOS inverter — or the CMOS inverter, in short. This is certainly the most popular at present and therefore deserves our special attention.

Principle of Operation

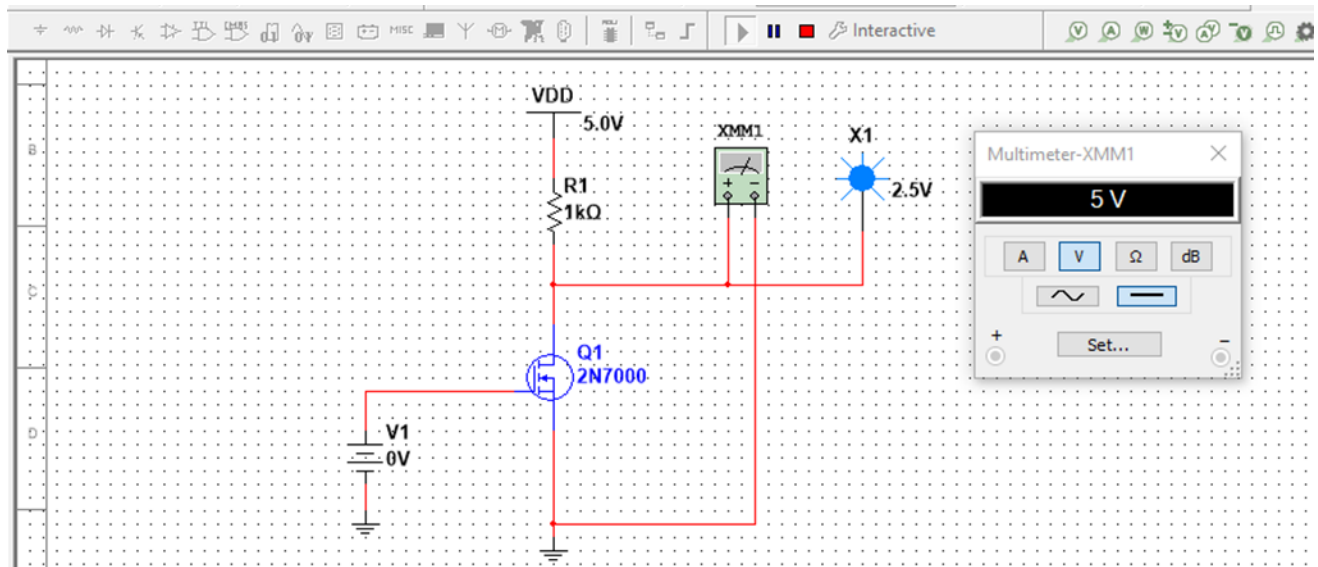
The logic symbol and truth table of ideal inverter is shown in figure given below. Here A is the input and B is the inverted output represented by their node voltages. Using positive logic, the Boolean value of logic 1 is represented by Vdd and logic 0 is represented by 0. Vth is the inverter threshold voltage, which is $V_{dd} / 2$, where Vdd is the output voltage.

The output is switched from 0 to Vdd when input is less than Vth. So, for $0 < V_{in} < V_{th}$ output is equal to logic 0 input and $V_{th} < V_{in} < V_{dd}$ is equal to logic 1 input for inverter.



2. Why NMOS transistor is selected as pull-down transistor?

A pull-down transistor should bring output to "0" means it should provide a path to the charges to go into ground terminal. NMOS transistor is selected as pull-down transistor because it provides path to ground without any threshold voltage drop i.e. V_T .



It is good for LOW pass logic “0”.

3. Why PMOS is connected to VDD?

In these devices, gate is reverse biased. PMOS consist of p-channel and to reverse bias the gate, a negative voltage is applied between gate and source. Therefore, PMOS (source terminal) is connected to VDD.

4 Which is better PMOS OR NMOS?

NMOS is better than PMOS because it has advantage of speed over PMOS because mobility of carriers (electrons) is greater than that of holes in PMOS.

Conclusion:

Any combinational logic circuit can be designed using NMOS transistors as a Pull-down network (as they provide path to ground when they are turned ON) and a resistor connected to VDD. NMOS transistors has advantage that they provide high speed as compared to PMOS transistors because they have mobility of electrons causing increase in the conduction.



INTEGRATED ELECTRONICS # 05

Design and analyze the behavior of CMOS as Inverter.

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LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	DESIGN AND CODING	OBSERVATION/ RESULTS	SCORE

Performance:

Objective: To be familiar with the CMOS operation as inverter Circuit.

Apparatus:

- P-MOS & N-MOS transistors
- Multimeter
- Connecting wire
- LED

DISCUSSION :

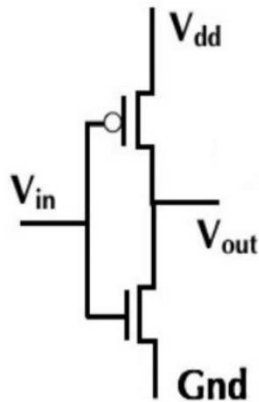
The inverter circuit as shown in the figure below. It consists of PMOS and NMOS FET. The input A serves as the gate voltage for both transistors.

The NMOS transistor has input from Vss (ground) and the PMOS transistor has input from Vdd. The terminal Y is output. When a high voltage ($\sim V_{dd}$) is given at input terminal (A) of the inverter, the PMOS becomes an open circuit, and NMOS switched OFF so the output will be pulled down to Vss.

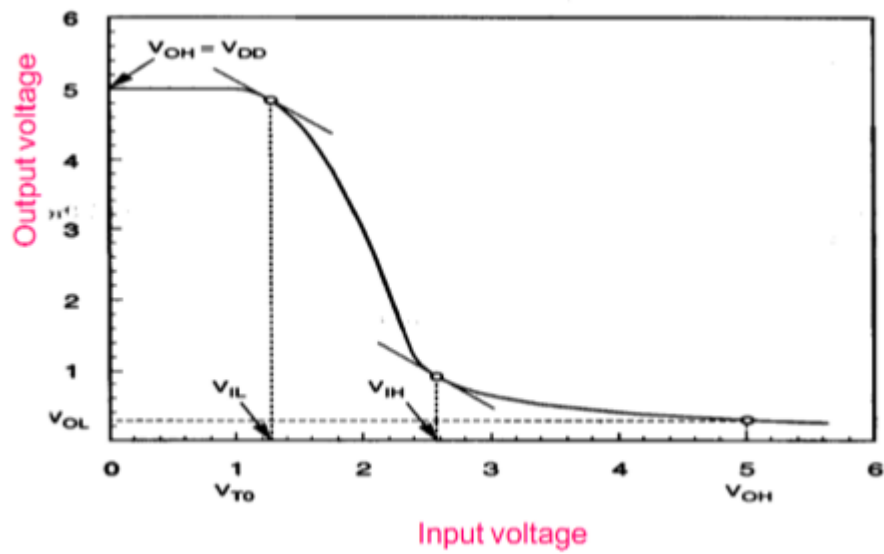
It consists of PMOS and NMOS FET. The input A serves as the gate voltage for both transistors. ... When a low-level voltage ($<V_{dd}$, $\sim 0\text{V}$) applied to the inverter, the NMOS switched OFF and PMOS switched ON. So the output becomes V_{dd} or the circuit is pulled up to V_{dd}

For the investigation of circuit-level degradation a CMOS (complementary MOS) inverter is analyzed. A major advantage of CMOS technology is the ability to easily combine complementary transistors, n-channel and p-channel, on a single substrate.

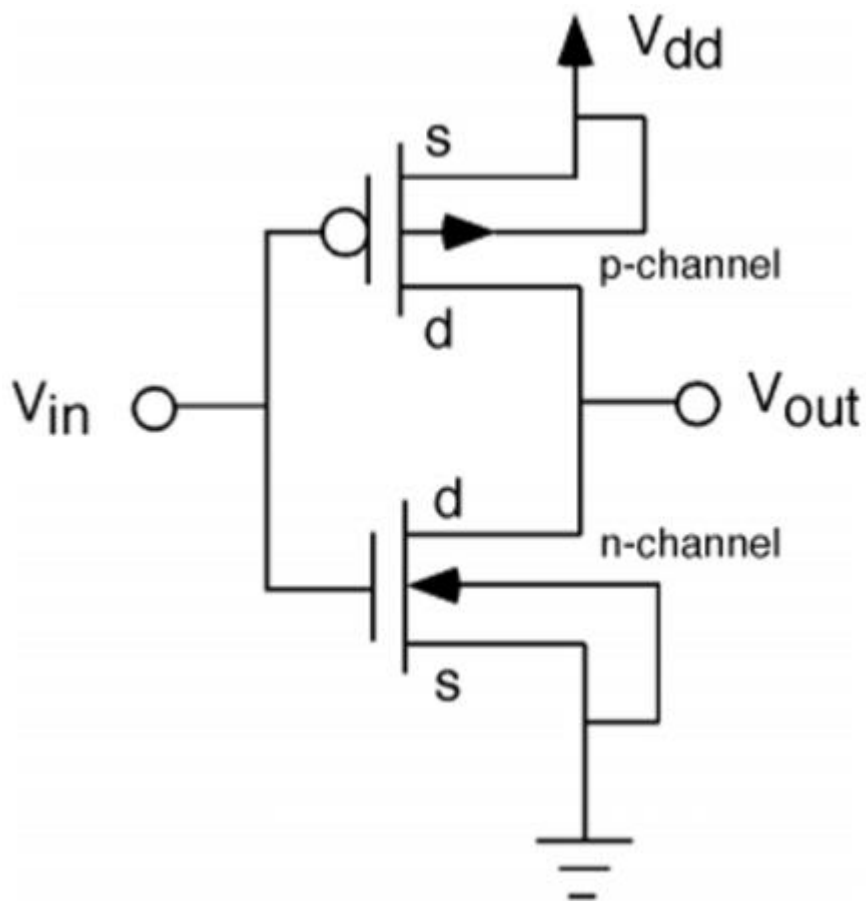
CIRCUIT DIAGRAM:



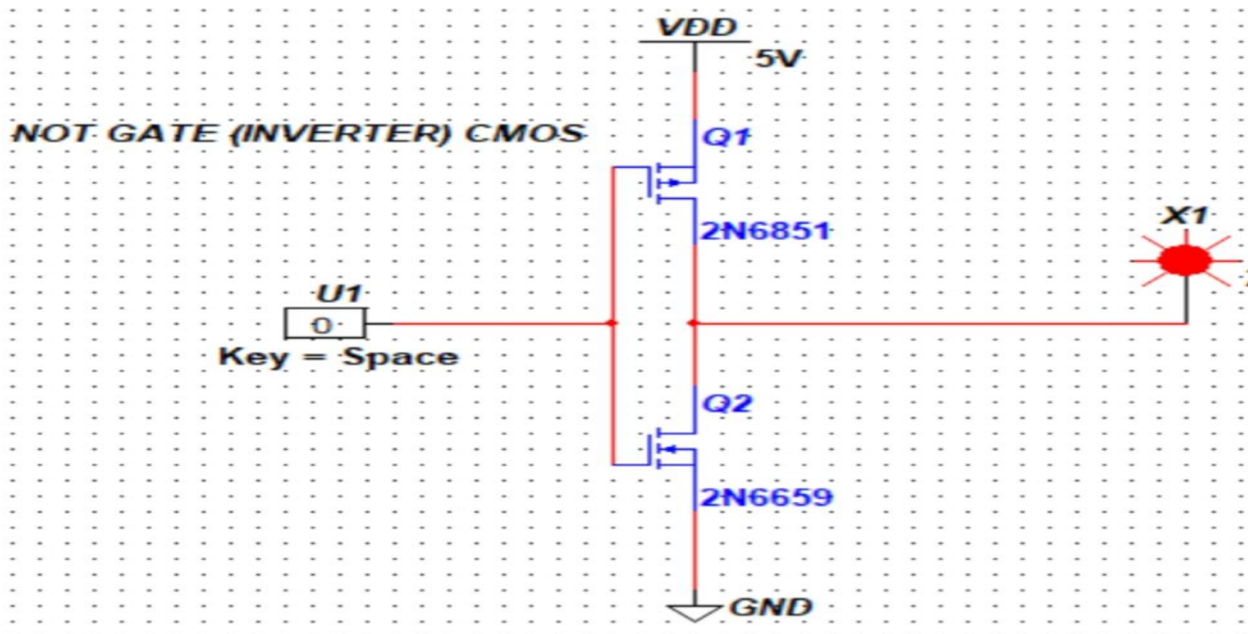
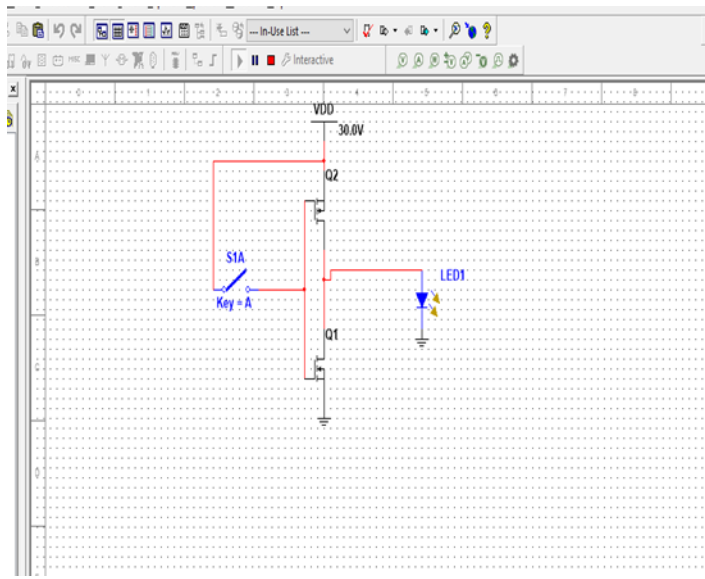
INPUT	LOGIC INPUT	OUTPUT	LOGIC OUTPUT
0 v	0	V_{dd}	1
V_{dd}	1	0 v	0



Detailed circuit diagram for CMOS Inverter



SIMULATION: design of cmos inverter



Review Questions

1. Distinguish between bipolar transistor inverter and mos inverter?

Bipolar Technology	CMOS
• Higher switching speed • It offers high current drive per unit area and high gain • Generally better noise performance and better high frequency characteristics • It has better analogue capability compare to others. • Improved I/O speed. • It offers high power dissipation. • lower input impedance (high drive current) • low packing density. • low voltage swing logic. • It offers lower delay sensitivity to load. • high gm ($gm \propto V_{in}$) • It offers high unity gain bandwidth at low current • They are basically unidirectional devices.	Following are the characteristics or benefits of CMOS technology: • It offers high noise margins. • It has low static power dissipation compare to other technologies. • It offers high packing density • It offers high yield along with large integrated complex functions. • It has low manufacturing cost per device. • Scalable threshold voltage • It has High input impedance and low drive current. • It offers higher delay sensitivity to load (i.e. it has fan out limitation.) • It has lower trans-conductance, here trans-conductance $gm \propto V_{in}$ • It has lower output drive current (This will have issue while driving higher capacitive loads) • A near ideal switching device • Bi-directional capability (drain & source are interchangeable)

Lab Task-1: CMOS Inverter Gate

- Connect the circuit shown in Figure VDD = 5V and value of resistor by your choice.
- Complete the following table.

V_{IN}	V_{OUT}
0	5V
5	220.189mV

- Complete the following table and use your readings to plot the voltage transfer characteristic (VTC) of Inverter logic gate.

V_{in}	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
V_{out}	5V	5V	5V	5V	5V	3.741V	676.676mV	399.45mV	305mV	253.4mV	220.189mV

- Determine V_{OH} , V_{OL} , V_{IH} , V_{IL}

